MEDICAL PARASITOLOGY — EXAM QUESTION BANK

Classified by Parasite Group & Disease (Direct Questions, True/False, Cases & MCQs with Model Answers)

Compiled From: 5 Past Exam & Quiz Models | **Organization:** By Taxonomic Group & Disease | **Includes:** Cestodes, Trematodes, Nematodes, Protozoa, Arthropods

1. General Parasitology & Introduction (مقدمة وعام)

I. DIRECT & ESSAY QUESTIONS (الأسئلة المقالية والمباشرة)

1. Talk about types of Symbiotic Association?

- **A. Mutualism:** A relationship in which both partners benefit from the association (e.g., blood-sucking leeches harboring intestinal bacterial species to digest blood; at least 20% of insects, mites, crustaceans, and filarial nematodes mutually infected with *Wolbachia* bacteria).
- **B. Commensalism:** One partner benefits from the association, while the host is neither helped nor harmed (e.g., commensal intestinal protozoa in humans such as *Entamoeba dispar*, *Endolimax nana*, and *Iodamoeba butschlii*).
- **C. Parasitism:** One participant (the parasite) harms or lives at the expense of the host, causing mechanical injury (boring/digging into tissues) or stimulating a damaging inflammatory/immune response.

2. What are the effects of parasites on their hosts?

- **Consumption of host's food:** e.g., *Diphyllobothrium latum* absorbs Vitamin B12 causing megaloblastic anemia.
- **Feeding on host's blood:** e.g., *Ancylostoma duodenale* and *Plasmodium* causing anemia.
- **Damage to host tissues:** e.g., *Fasciola hepatica* in the liver or *Entamoeba histolytica* flask-shaped ulcers.
- **Mechanical obstruction:** e.g., *Ascaris lumbricoides* blocking the intestine/bile duct, or *Wuchereria bancrofti* blocking lymphatics.
- **Toxic effects (poisons & secretions):** Causing severe inflammation or anaphylactic shock (e.g., ruptured Hydatid cyst fluid).
- **Sexual changes & Activation of growth:** Some parasites destroy host reproductive organs or induce gigantism in intermediate hosts (e.g., snails).

3. Talk about ways parasites enter their hosts (Modes of Entry)?

- **Mouth (Fecal-oral / Foodborne):** Ingestion of contaminated food, water, or undercooked meat (e.g., *Ascaris*, *Entamoeba*, *Taenia*).
- **Skin Penetration:** Direct larval penetration (e.g., *Ancylostoma* filariform larva, *Schistosoma* cercaria) or Arthropod vector bite (e.g., *Plasmodium* via *Anopheles* mosquito).
- **Inhalation:** Airborne eggs (e.g., *Enterobius vermicularis*).
- **Congenital (Transplacental):** Mother to fetus (e.g., *Toxoplasma gondii* tachyzoites).
- **Sexual Contact:** Direct venereal transmission (e.g., *Trichomonas vaginalis* trophozoite).
- **Blood Transfusion / Organ Transplant:** (e.g., *Plasmodium*, *Trypanosoma*, *Toxoplasma*).
- **Autoinfection:** Internal or external reinfection (e.g., *Strongyloides stercoralis*, *Hymenolepis nana*, *Taenia solium*, *Enterobius vermicularis*, *Cryptosporidium parvum*).

4. Define the following terms: Protozoa, TDS/TDH, PAM, Cyclodevelopmental transmission?

- **Protozoa:** Unicellular eukaryotic microorganisms capable of performing all vital physiological functions within a single cell.

- **TDS (Trichuris Dysentery Syndrome) / TDH:** Heavy infection with *Trichuris trichiura* leading to chronic mucoid/bloody diarrhea, rectal prolapse, tenesmus, and iron-deficiency anemia.
- **PAM (Primary Amoebic Meningoencephalitis):** A fulminant, rapidly fatal CNS infection caused by the free-living thermophilic amoeba *Naegleria fowleri*, acquired when contaminated warm freshwater enters the nasal cavity.
- **Cyclodevelopmental transmission:** Biological transmission in which the parasite undergoes developmental/morphological maturation inside the arthropod vector without multiplication in number (e.g., microfilariae in mosquitoes).

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. Life cycles of parasites may be heterogeneous (heteroxenous / indirect) e.g.: Repeated x2

a) *Fasciola spp.* | b) *Ascaris lumbricoides* | c) *E. vermicularis* | d) *H. nana*

Answer: a) *Fasciola spp.* (Requires multiple hosts to complete its life cycle).

2. Feed on the blood of the host: Repeated x2

a) *Ancylostoma* | b) *Plasmodium* | c) *Trypanosoma* | d) All of the above

Answer: d) All of the above

3. A relationship in which both partners benefit from the association: Repeated x2

a) Mutualism | b) Commensalism | c) Parasitism | d) All of the above

Answer: a) Mutualism

4. Adult is considered the infective stage in: Repeated x3

a) *Trichomonas vaginalis* | b) *Ancylostoma* | c) *Strongyloides stercoralis* | d) *Trypanosoma*

Answer: a) *Trichomonas vaginalis* (Exists only as an adult trophozoite stage; no cyst form).

5. Parasites may be transmitted by autoinfection as the following, EXCEPT: Repeated x4

a) *H. nana* | b) *T. solium* | c) *T. saginata* | d) *Strongyloides stercoralis*

Answer: c) *T. saginata* (*T. saginata* eggs are not infective to humans; they require cattle).

6. Praziquantel is the drug of choice given orally for the following worms:

a) *E. vermicularis* | b) *T. saginata* | c) *A. lumbricoides* | d) *T. trichiura*

Answer: b) *T. saginata* (Also in the "EXCEPT" version between *E. vermicularis, T. saginata, H. nana, T. solium*, the exception is **a) *E. vermicularis***).

7. Albendazole is the drug of choice given orally in a single dose of 400 mg for the following EXCEPT:

a) *E. vermicularis* | b) *Ancylostoma duodenale* | c) *H. nana* | d) *Ascaris*

Answer: c) *H. nana* (Treated with Praziquantel).

2. Cestodes / Tapeworms (الديدان الشريطية)

A. Taenia saginata & Taenia solium (Taeniasis & Cysticercosis)

I. DIRECT QUESTIONS

1. Compare between *Taenia saginata* and *Taenia solium* in terms of morphology, life cycle, and pathogenesis?

Feature	*Taenia saginata* (Beef Tapeworm)	*Taenia solium* (Pork Tapeworm)
Morphology	Length: 4–10 m (1000–2000 proglottids). Scolex: Pear-shaped, 4 suckers, **no rostellum/ hooks**. Gravid segment: **15–30** lateral uterine branches.	Length: 3–8 m (~1000 proglottids). Scolex: Globular, 4 suckers + **armed rostellum with double crown of hooks**. Gravid segment: **7–13** lateral uterine branches.
Life Cycle	Definitive host: Man. Intermediate host: **Cattle**. Infective stage to man: ***Cysticercus bovis***.	Definitive host: Man. Intermediate host: **Pig** (and accidentally **Man**). Infective stage: ***Cysticercus cellulosae*** (causes taeniasis) or **Eggs** (causes human cysticercosis).
Pathogenesis	Mild GI disturbance, abdominal discomfort, active crawling of single gravid segments out of the anus; rare intestinal obstruction.	Mild intestinal symptoms from adult worm (rarely causes obstruction). **Major hazard: Cysticercosis / Neurocysticercosis** (epilepsy, hydrocephalus, ocular damage) if man ingests eggs.

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. *Cysticercus bovis* is the infective stage of: Repeated x3

a) *E. granulosus* | b) *T. saginata* | c) *Fasciola spp.* | d) *T. solium*

Answer: b) *T. saginata*

2. All of the following are major symptoms of adult pork tapeworm (*T. solium*) infection EXCEPT: Repeated x2

Version 1: a) Intestinal obstruction | b) Abdominal pain | c) Eosinophilia | d) Epilepsy and CNS disorder
Version 2: a) Intestinal obstruction | b) Abdominal pain | c) Diarrhea | d) Weight loss

Answer: In Version 1: **d) Epilepsy and CNS disorder** (caused by larval cysticercosis, not adult tapeworm) or **a) Intestinal obstruction**. In Version 2: **a) Intestinal obstruction** (*T. solium* rarely causes obstruction according to the lecture text).

3. Which of the following statements best explains why eating pork does NOT necessarily lead to human cysticercosis?

A) Humans reproduce cysticerci | B) Humans ingest eggs and act as intermediate hosts | C) The adult worm is only in pork | D) Humans get infected by cercarial penetration

Answer: B) Humans ingest eggs and act as intermediate hosts (Eating undercooked pork containing *Cysticercus cellulosae* causes adult intestinal taeniasis, whereas cysticercosis requires ingesting *T. solium* eggs).

4. On examination of stool for eggs of *T. saginata* and *T. solium*, we cannot distinguish them from each other; differentiation depends on:

A) Examination of egg color | B) Gravid uterine branches and proglottid/scolex morphology | C) Egg pH | D) Egg operculum

Answer: B) Gravid uterine branches and proglottid/scolex morphology

5. A patient infected with cysticercosis without eating pork meat (or presenting with neurological damage having never eaten pork) acquired it by: Repeated x2

A) Ingestion of *Cysticercus cellulosae* | B) Ingestion and digestion of *Taenia solium* eggs | C) Ingestion of pig meat | D) Ingestion of eggs of *T. saginata*

Answer: B) Ingestion and digestion of *Taenia solium* eggs (via contaminated food/water or autoinfection).

6. In a patient with neurocysticercosis, the lesions (cysticerci) develop in different stages. What is the underlying mechanism?

A) Repeated exposure to eggs of *Taenia solium* | B) Multiplication of adult worms in the intestine | C) Transformation of cysticerci into proglottids | D) Re-infection by metacercariae

Answer: A) Repeated exposure to eggs of *Taenia solium*

7. Which combination is correct:

A) *T. solium* — cattle — cysticercosis | B) *T. saginata* — pig — cysticercosis | C) *T. solium* — pig — human cysticercosis | D) *T. saginata* — fish — hydatid cyst

Answer: C) *T. solium* — pig — human cysticercosis

8. Patient has intestinal *Taenia* infection but no tissue lesions; which stage represents the parasite present?

A) Cysticercus | B) Oncosphere | C) Adult tapeworm | D) Hydatid cyst

Answer: C) Adult tapeworm

9. Patient acquired taeniasis after eating undercooked beef by ingesting:

A) Eggs | B) Cysticerci (*Cysticercus bovis*) | C) Metacercariae | D) Oncospheres

Answer: B) Cysticerci

B. Echinococcus granulosus (Hydatid Disease / داء الأكياس المائية)

I. DIRECT & IDENTIFICATION QUESTIONS

1. Talk about the Life cycle and mode of transmission for *Echinococcus granulosus*?

- **Hosts:** Definitive hosts are dogs and other canines; Intermediate hosts are herbivores (sheep, cattle, goats, camels) and accidentally **humans** (dead-end host).
- **Mode of Transmission to Humans:** Ingestion of food, water, or raw vegetables contaminated with *E. granulosus* eggs excreted in dog feces, or direct hand-to-mouth transfer after handling infected dogs.
- **Life Cycle:** Adult worm inhabits the small intestine of dogs and releases embryonated eggs in feces $ ightarrow$ Intermediate host (sheep or man) ingests eggs $ ightarrow$ Oncosphere hatches in the duodenum, penetrates the intestinal wall, and enters the portal venous circulation $ ightarrow$ Lodges in the liver (60–70%), lungs (20%), brain, or bones, where it develops slowly into a fluid-filled **Hydatid cyst** containing brood capsules and protoscoleces $ ightarrow$ Dogs become infected by eating sheep viscera containing fertile hydatid cysts.

2. [Diagram Identification] Write the name of this parasite and identify parts 1 to 5? Repeated x2

Name: *Echinococcus granulosus* (Adult worm).
Indicated Parts: 1. Scolex (with 4 suckers and armed rostellum) | **2.** Short Neck | **3.** Immature segment (proglottid) | **4.** Mature segment | **5.** Gravid segment (filled with eggs).

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. Man is the definitive host for the following Tapeworms EXCEPT: Repeated x3

a) *T. saginata* | b) *E. granulosus* | c) *H. nana* | d) *D. latum*

Answer: b) *E. granulosus* (Dogs are the definitive host; man is an accidental intermediate host).

2. The PAIR method for treatment is used for: Repeated x4

a) Hydatid cyst | b) Ectopic fascioliasis | c) Chronic schistosomiasis | d) Cysticercosis

Answer: a) Hydatid cyst (Puncture, Aspiration, Injection of scolicidal agent, Re-aspiration).

C. Diphyllobothrium latum (Broad Fish Tapeworm)

I. DIRECT QUESTIONS

1. Compare between *Taenia saginata* and *Diphyllobothrium latum* in terms of morphology, life cycle, and pathogenesis?

Feature	*Taenia saginata*	*Diphyllobothrium latum*
Morphology	Scolex is quadrate with **4 cup-like suckers**; mature/gravid segments are longer than broad; uterus has lateral branches; non-operculated eggs.	Longest tapeworm (up to 10 m+); scolex is spatulate (spoon-shaped) with **2 longitudinal sucking grooves (bothria)**; segments are **broader than long**; rosette-shaped uterus; **operculated eggs**.
Life Cycle	Requires 2 hosts: Man (definitive) and 1 intermediate host (Cattle — *Cysticercus bovis*).	Requires **3 hosts**: Man (definitive) and **2 intermediate hosts** (1st: Freshwater crustacean/*Cyclops* [procercoid]; 2nd: Freshwater fish [plerocercoid/sparganum]).
Pathogenesis	Mild GI irritation, crawling proglottids.	Competes with the host for **Vitamin B12**, causing **Megaloblastic (Pernicious) Anemia** and neurological symptoms.

II. TRUE / FALSE & MCQS

1. [True / False] *D. latum* requires three hosts to complete its life cycle. Repeated x2

Answer: True (√) — Definitive host (Man) + 1st Intermediate host (*Cyclops*) + 2nd Intermediate host (Freshwater fish).

2. [MCQ] Which of the following parasites has Crustacean & freshwater fish as intermediate hosts?

A) *Taenia saginata* | B) *Taenia solium* | C) *Fasciola hepatica* | D) *Diphyllobothrium latum*

Answer: D) *Diphyllobothrium latum*

3. Trematodes / Flukes (الديدان المثقوبة)

A. General Trematodes & Schistosoma spp. (Blood Flukes / البلهارسيا)

I. TRUE / FALSE QUESTIONS

1. *Schistosoma mansoni* inhabits the blood vessels (inferior mesenteric veins) of the large intestine. Repeated x2

Answer: True (√)

2. Trematodes or flukes are members of the Platyhelminthes and are generally cylindrical. Repeated x2

Answer: False (X)

Correction: Trematodes are generally **flat, leaf-shaped** worms (Nematodes are cylindrical; among Trematodes, only *Schistosoma* is cylindrical).

II. MULTIPLE CHOICE & PICTURE QUESTIONS

1. General characteristics of Blood flukes (*Schistosoma*) are the following EXCEPT: Repeated x3

a) Sexes are separate (dioecious) | b) They are cylindrical, other flukes have a flat shape | c) Male is broader and female is filiform and longer than male | d) Metacercaria is the infective stage

Answer: d) Metacercaria is the infective stage (Schistosomes have no metacercaria stage; the fork-tailed **Cercaria** is the infective stage).

2. Schistosomular migration and maturation cause: Repeated x2

a) Katayama fever | b) Allergic reactions | c) Eosinophilia | d) All of the above

Answer: d) All of the above

3. The Schistosomes constitute the second major category of flukes and include organisms with separate sexes:

A) Intestinal Flukes | B) Blood Flukes | C) Liver Flukes | D) All of the above

Answer: B) Blood Flukes

4. Usually inhabiting the blood vessels (superior mesenteric veins) of the small intestine:

A) *Schistosoma mansoni* | B) *Schistosoma japonicum* | C) *Fasciola hepatica* | D) Not all

Answer: B) *Schistosoma japonicum*

5. Patients with Schistosomiasis develop severe Portal Hypertension. What is the cause?

A) Mechanical obstruction due to adult worms | B) Granulomatous deposition around eggs in portal tracts | C) Direct damage to hepatocytes by worms | D) Damage to portal vein by adult worms

Answer: B) Granulomatous deposition of eggs (leading to Symmers' pipe-stem periportal fibrosis).

6. A patient from an endemic area has hepatosplenomegaly and portal hypertension. Stool examination reveals eggs with a prominent lateral spine. The most likely parasite is:

A) *Schistosoma haematobium* | B) *Schistosoma japonicum* | C) *Schistosoma mansoni* | D) *Fasciola hepatica*

Answer: C) *Schistosoma mansoni*

7. Schistosomiasis has been associated with co-infections by the following pathogens:

a) HIV and Hepatitis viruses | b) *Salmonella* | c) *Neisseria spp.* | d) *Plasmodium* species

Note: All four groups (HIV/Hepatitis, chronic *Salmonella*/*Neisseria*, and *Plasmodium*) are explicitly listed as associated co-infections in the course material.

8. [Picture Identification] Liver histology section showing cellular granuloma surrounding a trapped parasite ovum:

Diagnosis: Bilharzial (Schistosomal) Granuloma in the liver.

B. Fasciola hepatica (Sheep Liver Fluke / ديدان الكبد)

I. DIRECT QUESTIONS

1. What is Ectopic Fascioliasis?

Answer: Ectopic fascioliasis occurs when migrating immature *Fasciola* larvae in the peritoneal cavity or bloodstream deviate from their normal route to the liver and settle in abnormal extra-hepatic tissues such as the subcutaneous tissue (causing migratory **eosinophilic panniculitis**), lungs (**pulmonary infiltrates**, pleuropericarditis), brain/meninges (**meningitis**), or lymph nodes (**lymphadenopathy**).

II. TRUE / FALSE QUESTIONS

1. *Lymnaea truncatula* snail is the intermediate host for *Fasciola hepatica*.

Answer: True (√)

2. Halzoun can cause pulmonary infiltrate, meningitis, and lymphadenopathy.

Answer: False (X)

Correction: Ectopic fascioliasis causes pulmonary infiltrate, meningitis, and lymphadenopathy. (Halzoun is acute pharyngeal fascioliasis caused by eating raw infected liver, leading to pharyngeal edema, dysphagia, and suffocation).

III. MULTIPLE CHOICE QUESTIONS (MCQS)

1. Ectopic fascioliasis can cause: Repeated x2

a) Pulmonary infiltrate | b) Meningitis and lymphadenopathy | c) Eosinophilic panniculitis | d) All of the above

Answer: d) All of the above

2. The larval stage for many parasites needs to encyst (on aquatic plants) to become the infective stage: Repeated x3

a) *Fasciola hepatica* (written as typo *E-hepaticis / E. histolytica* in some exams) | b) *S. haematobium* | c) *Ancylostoma* | d) *Ascaris*

Answer: a) *Fasciola hepatica* (Encysts as metacercariae on watercress/aquatic vegetation).

3. A patient presents with acute hepatic migration (e.g., acute fascioliasis). Which laboratory finding is expected?

A) Severe eosinophilia | B) Neutrophilia | C) Monocytosis | D) Lymphocytosis

Answer: A) Severe eosinophilia

C. Heterophyes heterophyes & Paragonimus westermani

MULTIPLE CHOICE QUESTIONS (MCQS)

1. The resulting watery diarrhea seems to be due to poor absorption of intestinal secretions from secretory crypt cells: Repeated x3

a) *H. heterophyes* | b) *Schistosoma haematobium* | c) *E. vermicularis* | d) None of the above

Answer: a) *Heterophyes heterophyes*

2. A patient presents with chronic cough & hemoptysis; which of the following is true for diagnosis (*Paragonimus westermani*)?

A) Operculated eggs in sputum | B) Non-operculated eggs in blood | C) Detection of eggs in urine | D) Trophozoites in sputum

Answer: A) Operculated eggs in sputum (and in stool if sputum is swallowed).

3. Type of trematode that directly presents with primary pulmonary disease:

A) *Fasciola* | B) *Schistosoma mansoni* | C) *Paragonimus westermani* | D) Not all

Answer: C) *Paragonimus westermani* (Lung fluke; note that one student key marked D by mistake, but medically *P. westermani* is the lung fluke).

4. Requires three hosts to complete its life cycle:

a) *Fasciola spp.* | b) *Paragonimus westermani* | c) *Schistosoma haematobium* | d) *H. nana*

Answer: b) *Paragonimus westermani* (Definitive: Man; 1st Intermediate: Snail; 2nd Intermediate: Crab/Crayfish).

4. Nematodes / Roundworms (الديدان الأسطوانية والخيطية)

A. General Characteristics of Nematodes

1. [True / False] Tissue nematodes are transmitted mainly by soil transmission, and most intestinal nematodes are fecal-oral route.

Answer: False (X)

Correction: Tissue nematodes are transmitted mainly by **insect vectors**, whereas intestinal nematodes are transmitted via the fecal-oral route or soil transmission.

2. [MCQ] General characteristics of intestinal nematodes include all of the following EXCEPT:

a) Non-segmented cylindrical or round worms | b) Sexes are separate with male smaller than female | c) Females are viviparous | d) Live in the human intestinal tract

Answer: c) Females are viviparous (Most intestinal nematodes are oviparous; Filarial tissue nematodes are viviparous).

B. Ascaris lumbricoides (Giant Roundworm / الأسكارس)

I. TRUE / FALSE & GIVE REASONS

1. [True / False] Eosinophilia is marked in adult ascariasis. Little or absent in larval ascariasis. Repeated x2

Answer: False (X)

Correction: Eosinophilia is marked in **larval ascariasis** (during hepatic/pulmonary migration) and little or absent in adult intestinal ascariasis.

2. [Give Reason] Eosinophilia is marked in larval ascariasis.

Answer: Because migrating larvae invade host tissues (liver and lungs, causing Loeffler's syndrome/pneumonitis), provoking a strong systemic hypersensitivity and eosinophilic immune response.

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. Adult female *Ascaris* may wander due to: Repeated x3

a) Overcrowding | b) Fever and drugs (e.g., anesthetics) | c) Irritation of worms by spicy foods | d) All of the above

Answer: d) All of the above (In the "EXCEPT" version, the false reason is: *"A female Ascaris lays about 200,000 eggs per day"*).

2. All of the following are complications of heavy *Ascaris lumbricoides* infection (wandering adult worms) EXCEPT: Repeated x2

a) Ileus and appendicitis | b) Obstructive jaundice & biliary stones | c) Liver abscess & intestinal perforation | d) Rectal prolapse

Answer: d) Rectal prolapse (Rectal prolapse is a classic complication of heavy *Trichuris trichiura* infection).

3. Blocking of the common bile duct with obstructive jaundice is caused by: Repeated x2

a) Adult ascariasis | b) *E. vermicularis* | c) Larval ascariasis | d) All of the above

Answer: a) Adult ascariasis

C. Enterobius vermicularis (Pinworm / الدودة الدبوسية)

I. TRUE / FALSE & GIVE REASONS

1. [True / False] *E. vermicularis* eggs are detected from feces in only 15% of infected individuals.

Answer: False (X)

Correction: Detected in stool in only about **5%** of infected individuals (because females lay eggs on the perianal skin at night; diagnosed via NIH/Scotch tape perianal swab).

2. [True / False] Heavy Pinworm infection may cause rectal bleeding causing chronic iron-deficiency anemia.

Answer: False (X)

Correction: This describes heavy **Whipworm (*Trichuris trichiura*)** infection. Pinworm mainly causes nocturnal pruritus ani.

3. [Give Reason] Pinworms may cause problems such as inflammation of the vagina (vaginitis).

Answer: Because gravid female worms migrating out of the anus onto the perianal region at night can crawl into the female genital tract (vagina, uterus, fallopian tubes), causing vulvovaginitis, endometritis, or salpingitis.

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. It is more prevalent in children aged 5–14 years and is more common in women than men:

a) *Ancylostoma duodenale* | b) *Trichuris trichiura* | c) *Ascaris lumbricoides* | d) *Enterobius vermicularis*

Answer: d) *Enterobius vermicularis*

D. Trichuris trichiura (Whipworm / الدودة السوطية)

I. TRUE / FALSE & MCQS

1. [True / False] Heavy whipworm infection may cause rectal prolapse, bloody diarrhea, and iron-deficiency anemia.

Answer: True (√)

2. [MCQ] Heavy whipworm infection may cause: Repeated x2

a) TDS (Trichuris Dysentery Syndrome) | b) Loeffler's syndrome | c) Liver abscess | d) All of the above

Answer: a) TDS (Trichuris Dysentery Syndrome)

3. [MCQ] Capable of causing anemia due to malnutrition and blood loss from a friable colon, unrelated to direct blood ingestion by the parasite:

a) Pinworm | b) Whipworm | c) Hookworm | d) All of the above

Answer: b) Whipworm (*Trichuris trichiura*)

4. [MCQ] *Trichuris trichiura* has the following characters EXCEPT:

a) Whipworm eggs mature in soil then hatch into first-stage rhabditiform larvae in soil | b) A soil-transmitted helminth | c) Adults inhabit the large intestine mainly the caecum | d) Female worms lay about 1,000–7,000 eggs daily

Answer: a) Whipworm eggs mature in soil then hatch in soil (Eggs embryonate in soil, but hatch only inside the small intestine after ingestion).

E. Ancylostoma duodenale (Hookworm / الأنكلستوما)

I. GIVE REASONS & MCQS

1. [Give Reason] Hookworms cause microcytic hypochromic anemia and eosinophilia.

- **Microcytic hypochromic (iron-deficiency) anemia:** Caused by chronic blood loss as adult worms suck blood via buccal cutting plates and secrete anticoagulants, leaving bleeding mucosal ulcers when they move from site to site.
- **Eosinophilia:** Caused by the host immune response to larvae migrating through the skin and lungs.

2. [MCQ] Moderate infection will gradually produce an iron-deficiency anemia: Repeated x2

a) Roundworm | b) Hookworm infection | c) Whipworm infection | d) Not all

Answer: b) Hookworm infection

3. [MCQ] The worms move from place to place (facultative habit of feeding), leaving bleeding ulcers:

a) *Ancylostoma duodenale* | b) *Trichuris trichiura* | c) *Ascaris lumbricoides* | d) *Strongyloides stercoralis*

Answer: a) *Ancylostoma duodenale*

4. [MCQ] Infection occurs when larvae in soil penetrate the skin OR mucous membranes of mouth and pharynx when ingested through contaminated food/water:

a) Hookworm infection (*Ancylostoma duodenale*) | b) Strongyloidiasis | c) Bancroftian filariasis | d) Not all

Answer: a) Hookworm infection

F. Strongyloides stercoralis (Threadworm)

I. TRUE / FALSE, GIVE REASONS & MCQS

1. [True / False] In Autoinfection (Hyperinfection) syndrome strongyloidiasis, the larva may be found in sputum.

Answer: True (√)

2. [Give Reason] Strongyloidiasis can range from asymptomatic eosinophilia in immunocompetent individuals to severe, life-threatening disease in immunocompromised patients.

Answer: In immunocompetent hosts, the immune system restricts the autoinfective cycle. In immunocompromised patients (e.g., on corticosteroids or HTLV-1/AIDS), uncontrolled internal autoinfection leads to **Hyperinfection Syndrome** with massive larval dissemination to lungs, CNS, and other organs, carrying enteric bacteria into the bloodstream and causing fatal sepsis.

3. [MCQ] Usually man acquires infection by filariform larva penetrating the skin: Repeated x2

a) *Echinococcus granulosus* | b) *Schistosoma haematobium* | c) *Strongyloides stercoralis* | d) Not all

Answer: c) *Strongyloides stercoralis* (Note: *Schistosoma* penetrates skin via cercariae, not filariform larvae).

4. [MCQ] The free-living generation in soil is the basic life cycle of (able to live either as a free-living worm in soil or parasitic worm in man): Repeated x2

a) *S. haematobium* | b) *Strongyloides stercoralis* | c) *E. vermicularis* | d) *F. hepatica*

Answer: b) *Strongyloides stercoralis*

5. [MCQ] This parasite is notable for its ability to persist and replicate within the host through an auto-infective cycle for decades:

a) *Strongyloides stercoralis* | b) *Ancylostoma duodenale* | c) *Trichuris trichiura* | d) *Enterobius vermicularis*

Answer: a) *Strongyloides stercoralis*

G. Tissue / Filarial Nematodes (Wuchereria, Brugia, Loa loa, Onchocerca)

I. TRUE / FALSE QUESTIONS

1. Periodicity means the presence of microfilariae in the peripheral circulation at 4 PM.

Answer: False (X)

Correction: Periodicity means the surge of microfilariae in peripheral blood at a specific time of day (4 PM applies only to subperiodic/diurnal forms, whereas classic nocturnal periodicity peaks between **10 PM and 2 AM**).

2. The best method of diagnosis of onchocerciasis is demonstration of microfilariae in bloodless skin snips.

Answer: True (√)

3. Elephantiasis occurs in the chronic stage of lymphatic filariasis due to obstruction of lymphatic vessels by larvae.

Answer: False (X)

Correction: Due to obstruction of lymphatic vessels by **adult worms** (living or dead) and resulting granulomatous fibrosis.

4. More microfilariae are found in capillary blood collected from the ear lobe than from the fingertip.

Answer: True (√)

5. *Loa loa* has a loose sheath; the tail end is free of nuclei and has diurnal periodicity.

Answer: False (X)

Correction: Microfilariae of *Loa loa* are sheathed and diurnal, but their **tail is full of nuclei to the very tip** (a tail free of nuclei is characteristic of *Wuchereria bancrofti*).

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. These nematodes are thread-like, characterized by being viviparous: Repeated x2

a) *Wuchereria bancrofti* | b) *Brugia malayi* | c) *Loa loa* | d) All of the above

Answer: d) All of the above (Note: When options are *Elephantiasis, Loa loa, Onchocerciasis*, choose ***Loa loa*** because the other two are disease names, not nematode species).

2. The pathological changes caused by *W. bancrofti* are related to the presence of:

a) Living microfilariae | b) Living or dead adult worms (and growing juveniles) | c) Living worms only | d) Not all

Answer: b) Living or dead worms (Living microfilariae elicit no significant pathology in classic lymphatic lesions).

3. Accompanied by recurrent bouts of fever known as "elephantoid fever":

a) The prepatent period | b) Chronic stage | c) Incubation period | d) Acute inflammatory stage

Answer: d) Acute inflammatory stage

4. For detection of microfilariae of *W. bancrofti* with diurnal periodicity, blood is collected:

a) At 1 p.m. (or 4 p.m. daytime) | b) At 2 a.m. | c) At 10 p.m. | d) Between 10 p.m. and 2 a.m.

Answer: a) At 1 p.m. (Only daytime option among nocturnal choices).

5. Drug that has a lethal effect on adult worms of *W. bancrofti*: Repeated x2

a) Hetrazan (DEC) | b) Ivermectin | c) Suramin | d) Albendazole

Answer: c) Suramin

6. Elephantiasis affects primarily the legs below the knee, and arms below the elbow (distinguishing it from Bancroftian filariasis): Repeated x2

a) *Wuchereria bancrofti* | b) *Brugia malayi* | c) *Loa loa* | d) *Onchocerca volvulus*

Answer: b) *Brugia malayi*

7. *Onchocerca volvulus* is transmitted by the bite of the:

a) Deerfly (mango fly) | b) Blackfly (*Simulium spp.*) | c) *Culex* mosquito | d) Tsetse fly (*Glossina*)

Answer: b) Blackfly (*Simulium spp.*)

8. Transmitted by the bite of the deerfly, also known as mango fly (*Chrysops*):

a) *Onchocerca volvulus* | b) *Wuchereria bancrofti* | c) *Loa loa* | d) Not all

Answer: c) *Loa loa*

9. Microfilariae appear in the peripheral blood at a specific time (Periodicity):

a) Adaptation between parasite and intermediate host | b) Exact causes remain obscure | c) Larva is present in peripheral blood when the vector bites | d) All of the above

Answer: d) All of the above

5. Protozoa (الأوليات الطبية)

A. Entamoeba histolytica (Amoebiasis / الأميبا)

I. CLINICAL CASE & TRUE / FALSE

1. [Clinical Case Diagnosis] A 13-year-old male complained of frequent diarrhea, fever, malaise, and abdominal colic. Stool culture for pathogenic bacteria was negative. Parasitological examination of stool revealed mono-nucleated, irregularly-shaped organisms about 20 µm, containing RBCs in the cytoplasm. What is your diagnosis?

Diagnosis: Acute Intestinal Amoebiasis (Amoebic Dysentery) caused by ***Entamoeba histolytica*** **trophozoites** (hallmark feature: ingested RBCs / erythrophagocytosis).

2. [True / False] House flies may deposit their faeces containing pathogens on human food or drink, e.g., cysts of *Entamoeba histolytica*.

Answer: True (√) (Mechanical transmission).

II. MULTIPLE CHOICE QUESTIONS (MCQS)

1. The trophozoites are the growing, feeding, tissue-invading stage and move by pseudopods:

a) *E. histolytica* | b) *Giardia lamblia* | c) *Trichomonas hominis* | d) *Trichomonas vaginalis*

Answer: a) *Entamoeba histolytica*

2. The most affected areas by *E. histolytica* in the intestine are the following parts EXCEPT:

a) The caecum | b) The right and left colonic flexures | c) The sigmoid-rectal colon | d) Small intestine

Answer: d) Small intestine (Inhabits the large intestine).

3. This is the result of repeated chronic amoebic infections with fibrous tissue deposition and superadded bacterial infection:

a) Amoeboma (amoebic granuloma) | b) Pericardial amoebiasis | c) Amoebic colonic perforation | d) Not all

Answer: a) Amoeboma (amoebic granuloma)

4. Used for examining motile trophozoites from material aspirated or scraped from the base of ulcers:

a) Direct diagnosis | b) Wet mount preparations (using warm saline) | c) Sigmoidoscopy | d) All of the above

Answer: b) Wet preparations

B. Giardia lamblia & Trichomonas spp. (Intestinal, Oral & Urogenital Flagellates)

I. TRUE / FALSE & MCQS

1. [True / False] Giardiasis is one of the commonest causes of waterborne diarrhea outbreaks worldwide.

Answer: True (√)

2. [True / False] *Giardia* is considered one of the nosocomial (hospital/daycare-transmitted) infections.

Answer: True (√) (Easily transmitted person-to-person in hospitals, psychiatric wards, and nurseries due to low infectious dose and immediately infective cysts).

3. [MCQ] Duodenal aspiration (or Entero-test string test) for detection of *G. lamblia* reveals:

a) Cysts and Trophozoites | b) Trophozoites only | c) Cysts only | d) Not all

Answer: b) Trophozoites only

4. [MCQ] Induces malabsorption due to decreased intestinal absorption of fats (steatorrhea), fatty acids, and fat-soluble vitamins:

a) *Giardia lamblia* | b) *E. histolytica* | c) *Trichomonas hominis* | d) All of the above

Answer: a) *Giardia lamblia*

5. [MCQ] Finding the trophozoites in unstained or stained smear from a swab taken from the oral cavity (tartar/gums):

a) *Trichomonas vaginalis* | b) *Trichomonas hominis* | c) *Trichomonas tenax* | d) Not all

Answer: c) *Trichomonas tenax*

C. Leishmania & Trypanosoma (Blood & Tissue Flagellates)

I. DIRECT & CLINICAL CASE QUESTIONS

1. What are the clinical stages of Human African Trypanosomiasis (Sleeping Sickness)?

- **1. Trypanosomal Chancre:** Local hard, painful inflammatory nodule/ulcer at the site of the Tsetse fly (*Glossina*) bite.

- **2. Stage I (Haemolymphatic Stage):** Systemic invasion of blood and lymphatics characterized by irregular fever, headache, hepatosplenomegaly, and posterior cervical lymphadenopathy (**Winterbottom's sign**).
- **3. Stage II (Meningoencephalitic Stage):** Invasion of the CNS causing daytime somnolence (sleeping sickness), nocturnal insomnia, tremors, meningoencephalitis, coma, and death.

2. [Clinical Case Diagnosis] An African patient presented with enlarged cervical lymph nodes, hepatosplenomegaly, fever, and generalized weakness. He had a history of having an indurated painful swelling on his face before his complaints. What is your diagnosis?

> **Diagnosis: Human African Trypanosomiasis (Sleeping Sickness)** — History of **trypanosomal chancre** (indurated painful swelling) followed by Stage I haemolymphatic dissemination with **Winterbottom's sign** (enlarged cervical lymph nodes), fever, and hepatosplenomegaly.

II. TRUE / FALSE, MCQS & PICTURE DIAGNOSIS

1. [True / False] Formol-gel test (Aldehyde test) is a specific test for diagnosis of Visceral Leishmaniasis.

> **Answer: False (X)** (It is a **non-specific** screening test that detects hypergammaglobulinemia; specific tests include rK39 dipstick, DAT, IFA, ELISA, or splenic/bone marrow aspirate).

2. [True / False] Miltefosine is the first oral drug with demonstrated efficacy against kala-azar (visceral leishmaniasis).

> **Answer: True (√)**

3. [MCQ] The lesion is usually single and occurs on exposed surfaces such as hands and face, known as a "dry oriental sore":

a) *L. major* | b) *L. tropica* | c) *L. aethiopica* | d) Not all

> **Answer: b) *Leishmania tropica*** (*L. major* causes the wet rural sore).

4. [Picture Identifications in Leishmaniasis]:

- **Destruction of nasal septum / oral-nasal mucosa:** Mucocutaneous Leishmaniasis (**Espundia** — *L. braziliensis*).
- **Face covered with non-ulcerating macular/nodular lesions after kala-azar:** Post Kala-azar Dermal Leishmaniasis (**PKDL**).
- **Single ulcerated lesion on face or ear:** Cutaneous Leishmaniasis (**Oriental Sore / Chiclero's Ulcer**).

D. Plasmodium spp. (Malaria / الملاريا)

I. DIRECT QUESTIONS

1. Why is *Plasmodium falciparum* named malignant tertian malaria?

- **1. Invasion of RBCs of all ages:** Invades both young reticulocytes and mature erythrocytes, resulting in very high parasitemia (up to 30–50%+ of RBCs).
- **2. Multiple infection of single RBCs:** Multiple ring forms frequently infect a single erythrocyte.
- **3. Cytoadherence, Rosetting & Sequestration:** Infected RBCs express sticky surface knobs (PfEMP1) that adhere to deep capillary endothelium in vital organs (brain, kidneys, lungs, placenta), causing microvascular obstruction, tissue hypoxia, lactic acidosis, **Cerebral Malaria**, and **Blackwater Fever**.

II. TRUE / FALSE QUESTIONS

1. Homozygotes suffering from sickle cell disease (HbSS) are at increased risk of severe falciparum malaria, especially severe anemia.

> **Answer: True (√)** (Heterozygous sickle cell trait HbAS protects against severe *P. falciparum*, whereas homozygous HbSS patients suffer fatal hemolytic crises).

2. The malaria asexual cycle occurs in the female *Anopheles* mosquito, which can be considered the intermediate host.

> **Answer: False (X)**

> **Correction:** The **sexual cycle (sporogony)** occurs in the female *Anopheles* mosquito (**definitive host**), whereas the **asexual cycle (schizogony)** occurs in man (**intermediate host**).

3. In *P. malariae*, growing trophozoites and schizonts may be found in peripheral blood only in fatal conditions.

> **Answer: False (X)**

> **Correction:** This applies to ***Plasmodium falciparum*** (where mature trophozoites and schizonts are sequestered in deep capillaries and appear in peripheral blood only in severe/fatal infections).

III. MULTIPLE CHOICE & PICTURE QUESTIONS

1. *Plasmodium ovale* causes:

a) Benign tertian malaria (ovale malaria) | b) Sub-tertian malaria | c) Quartan malaria | d) Malignant malaria

> **Answer: a) Benign tertian malaria**

2. The malaria infective stage to man (via mosquito bite) is:

a) Gametocytes | b) Trophozoite | c) Sporozoites | d) Merozoites

> **Answer: c) Sporozoites**

3. Merozoites invade RBCs and feed on glucose and hemoglobin, depositing a brownish-black breakdown product in the food vacuole called:

a) Schuffner's dots | b) Lactic acidosis | c) Malarial pigment (Hemozoin) | d) All of the above

> **Answer: c) Malarial pigment (Hemozoin)**

4. Human malaria has no animal reservoir host EXCEPT:

a) *P. vivax* | b) *P. falciparum* | c) *P. malariae* | d) *P. ovale*

> **Answer: c) *P. malariae*** (Higher apes/monkeys act as reservoir hosts).

5. Some sporozoites remain dormant within liver hepatocytes (in *P. vivax* and *P. ovale*) causing true relapse; these are called:

a) Merozoites | b) Trophozoites | c) Hypnozoites | d) Schizonts

> **Answer: c) Hypnozoites**

6. All stages (ring, mature trophozoite, schizont, and gametocytes) can be identified in peripheral blood in all of the following EXCEPT:

a) *P. falciparum* | b) *P. malariae* | c) *P. vivax* | d) *P. ovale*

Answer: a) *P. falciparum* (Only ring stages and crescent/banana-shaped gametocytes are normally seen in peripheral blood).

7. *Plasmodium* in the female *Anopheles* mosquito is an example of which mode of transmission:

a) Propagative transmission | b) Mechanical transmission | c) Biological transmission (Cyclopropagative) | d) Not all

Answer: c) Biological transmission

8. [Picture Identification] Dark/cola-colored urine sample in a patient with severe falciparum malaria:

Diagnosis: Malarial Hemoglobinuria (Blackwater Fever) due to massive intravascular hemolysis.

E. Toxoplasma gondii (Toxoplasmosis / داء المقوسات)

I. TRUE / FALSE, MCQS & PICTURE DIAGNOSIS

1. [True / False] The mature oocyst of *Toxoplasma gondii* contains four sporocysts, each containing two sporozoites.

Answer: False (X)

Correction: The mature (sporulated) oocyst contains **two sporocysts**, each containing **four sporozoites**.

2. [True / False] No therapeutic agent shows activity against the encysted tissue form (bradyzoites) of toxoplasmosis.

Answer: True (√)

3. [MCQ] Two antimicrobial regimens typically used to reduce the risk of congenital toxoplasmosis during pregnancy:

a) Spiramycin and pyrimethamine-sulfadiazine plus folinic acid | b) Trimethoprim and sulfamethoxazole | c) Pyrimethamine plus clindamycin | d) All of the above

Answer: a) Spiramycin and pyrimethamine-sulfadiazine plus folinic acid

4. [MCQ] The definitive host (the cat) acquires *Toxoplasma* infection by eating:

a) Tissue cysts containing bradyzoites | b) Pseudocysts containing tachyzoites | c) Mature oocysts | d) All of the above

Answer: d) All of the above

5. [MCQ] In congenital transmission, blood transfusion, and acute infection, the rapidly multiplying infective stage is:

a) Tachyzoites | b) Bradyzoites | c) Mature oocysts | d) Gametocytes

Answer: a) Tachyzoites

6. [MCQ] Toxoplasmosis represents a life-threatening complication in:

a) Patients with AIDS | b) Pregnant women (fetus) | c) Organ graft recipients | d) All of the above

Answer: d) All of the above

7. [Picture Identification] Infant presenting with markedly enlarged head circumference:

Diagnosis: Congenital Toxoplasmosis (Hydrocephalus).

6. Medical Entomology / Arthropods & Myiasis (الحشرات الطبية والتدويد)

I. DIRECT QUESTIONS

1. Define Myiasis and classify it according to the anatomical site of tissue invaded?

- **Definition:** Infestation of live human or vertebrate organs and tissues by dipterous fly larvae (maggots) which, for at least a certain period, feed on the host's dead or living tissue, liquid body substances, or ingested food.
- **Anatomical Classification:**
  - **1. Internal Myiasis:** Intestinal myiasis, Urogenital myiasis.
  - **2. External Myiasis:** Cutaneous myiasis (Traumatic/wound, Furuncular, Creeping), Ocular myiasis (Ophthalmomyiasis), Nasopharyngeal myiasis, Aural myiasis (Otomyiasis).

2. Talk about the treatment of Myiasis?

- **1. Cutaneous / Furuncular Myiasis:** Occlusion/suffocation of the breathing spiracles using petroleum jelly (Vaseline), liquid paraffin, or heavy oil to force the larva to emerge, followed by careful surgical/mechanical extraction with sterile forceps (avoiding crushing the larva in situ to prevent severe foreign-body inflammatory reaction).
- **2. Traumatic / Cavity Myiasis:** Surgical debridement and removal of maggots under local anesthesia, irrigation with antiseptic solutions (e.g., chloroform in vegetable oil or ethanol), tetanus prophylaxis, and systemic antibiotics for secondary bacterial infection.
- **3. Intestinal / Systemic Therapy:** Administration of purgatives/laxatives, broad-spectrum antiparasitic therapy (oral **Ivermectin**), and supportive care.

II. MULTIPLE CHOICE & PICTURE QUESTIONS

1. The organism undergoes both developmental changes and multiplication inside the arthropod vector:

a) Propagative transmission | b) Cyclodevelopmental transmission | c) Cyclopropagative transmission | d) Mechanical transmission

Answer: c) Cyclopropagative transmission (e.g., *Plasmodium* in *Anopheles*).

2. Cyclodevelopmental transmission means:

a) Multiplication only | b) No change in morphology | c) Both multiplication and morphological change | d) No multiplication in number, but there is simply a change in morphology/development

Answer: d) No multiplication in number, but there is simply a change in morphology (e.g., Filaria).

3. Examples of diseases caused directly by Arthropods (as etiological agents, not vectors):

a) Allergic reactions | b) Malaria | c) Leishmaniasis | d) All of the above

Answer: a) Allergic reactions (Malaria and Leishmaniasis are transmitted by vectors, not caused by the arthropod itself).

4. Ocular, nasopharyngeal, and aural myiasis can be caused by:

a) *Sarcophaga* flies | b) *Calliphora* flies | c) *Musca domestica* flies | d) All of the above

Answer: d) All of the above (Along with *Lucilia* flies).

5. From the general (systemic) treatment of Scorpion sting:

a) Ligate the limb with a tourniquet proximal to the sting | b) Incise over the sting until blood washes toxin | c) Cover wound with potassium permanganate | d) Not all

Answer: d) Not all (Options a, b, and c are listed under *Local* measures in the lecture notes; *General* systemic treatment includes airway maintenance, blood pressure support, and specific antivenom).

6. [Picture Identifications in Myiasis]:

- **Boil-like nodular skin swelling with central pore and protruding larva: Furuncular Myiasis** (e.g., *Dermatobia hominis* / *Cordylobia anthropophaga*).
- **Open necrotic wound containing numerous maggots: Traumatic (Wound) Myiasis**.